

Answer Part 1

Git workflow and peer review

TokTickIT - Lab 3 release evidence

Supapanya Yathip | 67070503443

Partner reviewer: Atip Infa-Udom | Atip-Infa

MAIN RELEASE VERIFIED. Atip-Infa merged PR #54 on 2026-09-14. The implementation was tested at main commit f401341. Final documentation is prepared for peer review; Issue #41 remains open until that integration is complete.

Verified implementation: f40134124261d2861d7f480adec34f074561fc94.

[Repository](#) | [Project board](#) | [Final audit Issue #41](#)

Issue	Increment	PR to lab3-staging
32	Engineering contract	42
33	Migration and seed	43
34	Authentication API	44
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PRs #42-#51 were peer-merged into staging. Atip-Infa approved PR #51 at e0a9c75 and merged it as 75bd6d3.

PR #52 documentation was approved at 75d65ff and peer-merged as 4fc4859. [PR #52 review and responses](#).

PR #53 was approved at b3242ba and merged as e233f8e. PR #54 was approved at e233f8e and merged to main as f401341. [Release approval](#).

[PR #51 approval](#) | [Author after approval](#) | [Author after merge](#)

Nine reciprocal reviews of Atip-Infa PRs #37-#45, submitted and merged by zerotwobook, include both partner responses in [reviewer.md](#). Partner PR #41 has a recorded title/review versus Issue/branch mismatch. PR #42 in our repository approved an earlier head; its later merge is recorded separately.

Document links target main. This final-main evidence update is prepared on docs/41-final-main-evidence and awaits documentation review and integration.

Answer Part 2

Specification-driven development

The engineering contract preceded feature implementation in Issue #32 / PR #42. The student explicitly selected Admin user management and read-only Ticket access, with IT Staff performing Ticket mutations.

Contract area	Implemented boundary
Identity and roles	One role per user; real password login; mandatory initial-password change; inactive accounts denied.
Requester	Session-owned Tickets and files, public comments, advisory resolution indication; no formal resolution.
IT Staff	Queue, claim/reassign, IT Priority, eight-state workflow, public history/comments and private notes.
Administrator	User create/edit/activation/password reset. Ticket/comments/notes are read-only.
Data evolution	Preserve existing Ticket/Attachment identity; evolve Development Requesters into Users; repeatable guarded seed.
Excluded	No self-registration, email delivery, multiple roles, user deletion, bulk operations or Actions Taken.

Source contracts: [specification.md](#) | [api-spec.md](#) | [ui-spec.md](#) | [migration.md](#)

FR/BR/AC, exact API schemas, safe errors, role matrix, workflow transitions and Definition of Done remain linked to actual test files in [tests.md](#).

Answer Part 3

Test-driven development and verification

The plan was reviewed with the contract before implementation. Tests distinguish unit/component checks, real PostgreSQL API/concurrency checks, mocked browser checks and real-browser E2E.

Observed execution	Result and limits
Server regression	445/445 in 27 files, 114.80 s on 2026-09-14; no failures or skips. Includes the new matrix and safe-error suites.
Traceability correction	API-06: 195 role/method checks using real HTTP login. API-25: 16 safe-error cases. API-04/07 now link completed Admin/Requester tests.
Client regression	82/82 in 13 files, 54.97 s on verified main.
Real browser suite	14/14 passed in 1.0 min on verified main with queue, upload recovery, user safety and responsive additions.
Build	Client and server production builds passed; Prisma schema validation passed on verified main.
Long-text regression	Initially failed all three roles. Detail grid wrapping corrected; responsive tests then passed.
Mocked browser regressions	Queue, operations, communication and users: 3/3 each across desktop, tablet and mobile. These complement real E2E.
Final main	Passed at f401341. Node 24.19.0; npm 11.17.0; PostgreSQL 16.13. Documentation integration is separate.

Latest server command: **npm test --workspace server -- --maxWorkers=1**; build: **npm run build**. Client: **npm test --workspace client -- --maxWorkers=1**. All were rerun on released main.

Reproduce with TEST_DATABASE_URL set to the isolated local toktickit_lab3_test database and **npm run test:e2e:lab3**. Dedicated ports 3006/5176, owned random schema and temporary file storage prevent tests from reusing the working app. Existing database contents are not reset.

Matrix grants reach expected read/validation/lookup handlers; successful mutations are covered separately by the domain suites. Safe-error checks inject database/storage failures and verify JSON envelopes without sensitive data.

The upload recovery case aborts the first HTTP upload as deliberate fault injection. Retry, database persistence, download bytes and authorization use the real system. The expiry case changes only its owned session's persisted expiration.

Commands, file-level traceability and remaining limitations: [tests.md](#) and [Final evidence branch](#).

Answer Part 4

AI use and My Reflection

Assistant: OpenAI Codex (GPT-6). I use AI to summarize lab documents, help write and fix code, explain errors, and troubleshoot technical problems. It also helps organize documentation and the GitHub workflow.

These are concise English paraphrases of eight actual requests/answers, following the current [ai-use.md](#). Technical checks were assistant-run; my role included project decisions and workflow requirements.

#	Representative prompt	How AI helped / my decision
1	Read the Lab 3 sheet and project carefully before implementing.	Understand requirements and existing code before I authorized work.
2	Summarize Lab 3 and the Issues we need.	Organize ten work packages to guide implementation.
3	Read skill.md and remind me at each GitHub workflow step.	Apply branches, linked Issues, peer review and complete PR metadata.
4	Admin manages users and views Tickets; IT Staff edits Tickets.	Apply my chosen role separation in the contract and code.
5	Does the specification follow the original lab sheet?	Check alignment before I asked it to continue.
6	Find the code causing the error and help fix it.	Explain the cause, fix related code and rerun relevant checks.
7	Check what is still missing before continuing.	Audit tests, code and documents; explain remaining work.
8	Finish documents and ask me before going to main.	Keep a documentation checkpoint before release.

Critical-thinking. I clarified the Admin/IT Staff role decision. Real-browser checks later exposed a long-text layout problem; AI fixed it and reran the checks. This illustrates why the working application and test evidence matter alongside suggestions.

My Reflection

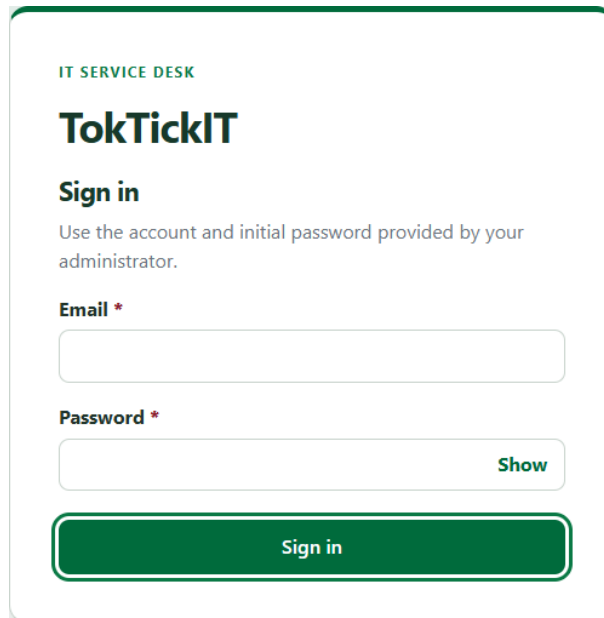
AI is useful for turning the lab sheet into a clearer summary and a sequence of tasks. As a specification assistant, it helps connect the requirements to the existing project and brings up unclear points for me to decide. This makes it easier to follow what needs to be built.

As a coding assistant, AI helps write and adjust code, explain errors, and troubleshoot technical problems involving Docker, PostgreSQL, Prisma, authentication, tests, and responsive layouts. It also helps organize the documentation and GitHub workflow. I still need to decide whether the proposed behavior matches the assignment and whether the documents describe the work the way I intend. For this lab, I asked to finish and check the documentation before moving to main.

Answer Part 5

Login and mandatory password change

Actual email/password login routes three roles to their permitted home. Invalid and inactive credentials produce the same safe failure. Password fields clear after failure. Initial/reset passwords require a change before protected API access. Logout and persisted expiry deny the old session.



IT SERVICE DESK

TokTickIT

Sign in

Use the account and initial password provided by your administrator.

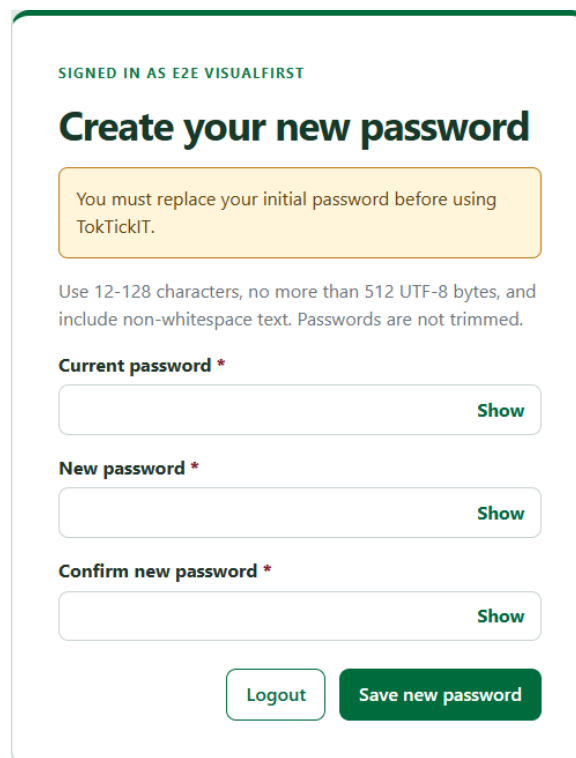
Email *

Password *

[Show](#)

[Sign in](#)

Real application capture: login-region.png (complete captured region, no image cropping).



SIGNED IN AS E2E VISUALFIRST

Create your new password

You must replace your initial password before using TokTickIT.

Use 12-128 characters, no more than 512 UTF-8 bytes, and include non-whitespace text. Passwords are not trimmed.

Current password *

[Show](#)

New password *

[Show](#)

Confirm new password *

[Show](#)

[Logout](#) [Save new password](#)

Real application capture: mandatory-password-region.png (complete captured region, no image cropping).

Answer Part 6

IT Staff Ticket Queue

The real queue supports combined search, status, owner and priority filtering, stable sorting, pagination and no-results recovery. The browser test verifies 12 matching rows split into pages of 10 and 2, ordered by Summary. Small screens use labelled cards.

TokTickITIT Service DeskE2E staffIT StaffChange PasswordLogout

Ticket Queue

Ticket Queue

All service requests across the support team.

Search

Ticket number, summary or requester

Status

All statuses

Category

All categories

Related System

All systems

Requested Priority

All priorities

IT Priority

All priorities

Owner

All owners

Sort by

Last updated

Order

Descending

Page size

10

Clear filters

Apply filters

Ticket Number	Created Date	Summary	Category	Requested Priority	IT Priority	Status	Owner	Last Updated
TKT-2026-000001	9/14/2026	Attachment persistence browser regression E2E requester / E2E Laptop	E2E Hardware	Medium	Medium	New	Unassigned	9/14/2026
TKT-2026-910011	9/14/2026	Queue audit 11 E2E requester / E2E Laptop	E2E Hardware	Low	Low	Open	Unassigned	9/14/2026
TKT-2026-910010	9/14/2026	Queue audit 10 E2E requester / E2E Laptop	E2E Hardware	Low	Low	Open	Unassigned	9/14/2026
TKT-2026-910009	9/14/2026	Queue audit 09 E2E requester / E2E Laptop	E2E Hardware	Low	Low	Open	Unassigned	9/14/2026
TKT-2026-910008	9/14/2026	Queue audit 08 E2E requester / E2E Laptop	E2E Hardware	Low	Low	Open	Unassigned	9/14/2026
TKT-2026-910007	9/14/2026	Queue audit 07 E2E requester / E2E Laptop	E2E Hardware	Low	Low	Open	Unassigned	9/14/2026
TKT-2026-910006	9/14/2026	Queue audit 06 E2E requester / E2E Laptop	E2E Hardware	Low	Low	Open	Unassigned	9/14/2026
TKT-2026-910005	9/14/2026	Queue audit 05 E2E requester / E2E Laptop	E2E Hardware	Low	Low	Open	Unassigned	9/14/2026
TKT-2026-910004	9/14/2026	Queue audit 04 E2E requester / E2E Laptop	E2E Hardware	Low	Low	Open	Unassigned	9/14/2026
TKT-2026-910003	9/14/2026	Queue audit 03 E2E requester / E2E Laptop	E2E Hardware	Low	Low	Open	Unassigned	9/14/2026

Showing 1-10 of 14 Tickets

PreviousNext

Real application capture: staff-home-desktop.png (complete captured region, no image cropping).

Priority/status badges now use the shared Zen styling. Queue query validation, unauthorized access and safe failures are also covered by API/component tests; see [staff-queue.md](#).

Answer Part 7

Ticket operations, communication and requester regression

Real E2E covers Requester creation, Staff search/claim/reassign/priority, public comments, internal notes, Requester indication, Staff resolve/close/reopen, and read-only Admin detail. Reopen clears obsolete resolution/closure/indication fields. Requesters cannot read private notes, another requester's Ticket or file.

Ticket operations

Claim Ticket

Assigned owner

Unassigned

Set IT Priority

Medium

Next status

Choose next status

Unassign owner

Save IT Priority

Change status

Public status reason (optional)
Visible to the requester. Required reasons must contain 5–1000 characters.

Public status history

No status changes yet.

Real application capture: staff-detail-region.png (complete captured region, no image cropping).

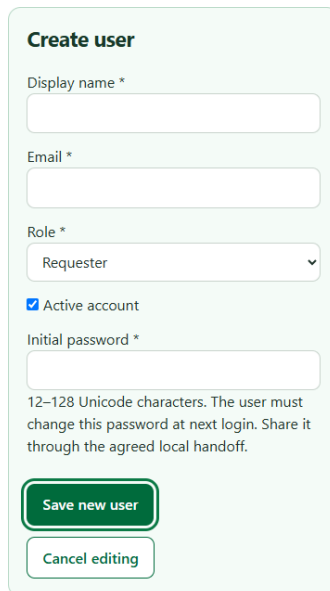
Requester regression checks a failed first upload and manual retry against the same saved Ticket, search, identical downloaded bytes after reload, cross-user denial, and removed-file denial after reload.

Detailed constraints and historical tests: [staff-operations.md](#) | [communication.md](#).

Answer Part 8

Administrator user management

The real browser flow creates a user, changes the initial password, edits the display name, resets the password, verifies old-session revocation and forced change, deactivates/reactivates the account and searches by name/role. Duplicate email is rejected; the current/last Administrator cannot be disabled through the UI. The API additionally enforces these guards against direct requests and concurrent changes.



Create user

Display name *

Email *

Role *

Requester

☒ Active account

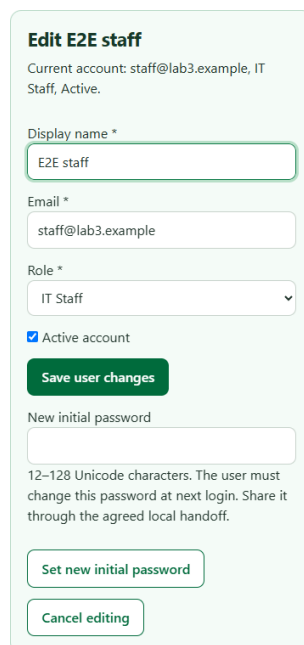
Initial password *

12–128 Unicode characters. The user must change this password at next login. Share it through the agreed local handoff.

Save new user

Cancel editing

Real application capture: admin-create-region.png (complete captured region, no image cropping).



Edit E2E staff

Current account: staff@lab3.example, IT Staff, Active.

Display name *

E2E staff

Email *

staff@lab3.example

Role *

IT Staff

☒ Active account

Save user changes

New initial password

12–128 Unicode characters. The user must change this password at next login. Share it through the agreed local handoff.

Set new initial password

Cancel editing

Real application capture: admin-edit-reset-region.png (complete captured region, no image cropping).

Answer Part 9

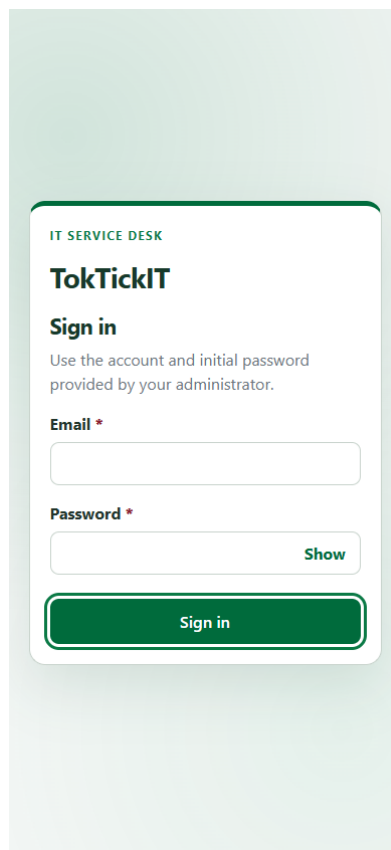
Zen Green, responsive layout and accessibility

Shared cards, fields, buttons, readable badges and role navigation continue the Lab 2 theme. The final audit checks 1440x900, 834x1112 and 390x844 CSS viewports and 720x450 reflow equivalent to 200% of a 1440x900 viewport; this is not an OS/browser chrome zoom assertion.

Description

[illegible]

Real application capture: requester-detail-region.png (complete captured region, no image cropping).



Real application capture: login-mobile.png (complete captured region, no image cropping).

Long persisted Description text wraps inside the detail grid. Keyboard checks cover Login tab order, create-user initial password before Save, and Staff dialog Escape/focus restoration. Full screenshot inspection/checklist is recorded separately in [ui-spec.md](#).

Release evidence: PR #54 peer approval and main merge are recorded; final-main automated checks passed on 2026-09-14. Final documentation review and board completion remain before submission. The older DRAFT PDF is historical; use this report for final documentation review.